
ORBITS

NOTES ON CALCULATIONS ON CLASSICAL ORBITS UNDER GRAVITY

Hannah Michelle Ellis

November 19, 2025

Contents

1	Plane Polar Coordinate System	2
1.1	Polar unit vectors	2
1.2	Velocity and acceleration in polar coordinates . . .	4
1.2.1	Position	4
1.2.2	Velocity	4
1.2.3	Acceleration	4
2	Gravity	6
2.1	Gravity as a central force	6
3	Angular Momentum	7
3.1	Conservation Of Angular Momentum	7
4	Energy	9
4.1	Potential Energy	9
4.2	Kinetic Energy	9
4.2.1	Total Energy	10
4.3	Conservation of Total Energy	10
5	Elliptical Orbits	12
5.1	Terminology	12
5.2	Solving equtaions of motion	12
5.3	Solution	14

List of Tables

List of Figures

1	Position and unit vectors in plane polar coordinates	2
---	--	---

1 Plane Polar Coordinate System

In this section we will derive the equations of motion for an object acted upon by gravity in plane polar coordinates. To do this we will first consider plane polar coordinates themselves and the derivatives of the position vector within the coordinate system.

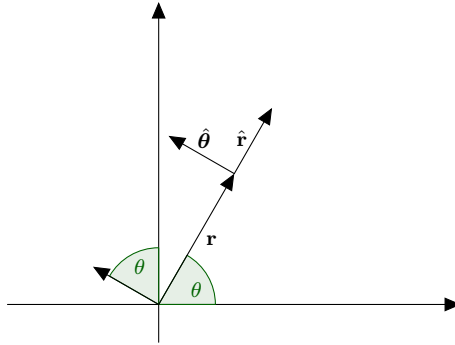


Figure 1: Position and unit vectors in plane polar coordinates

1.1 Polar unit vectors

The unit vectors of plane polar coordinates can be determined from the standard unit vectors as

$$\hat{\mathbf{r}} = \hat{\mathbf{i}} \cos(\theta) + \hat{\mathbf{j}} \sin(\theta) \quad (1.1)$$

$$\hat{\boldsymbol{\theta}} = -\hat{\mathbf{i}} \sin(\theta) + \hat{\mathbf{j}} \cos(\theta) \quad (1.2)$$

Where r is the distance from the origin to the position or alternatively the length of the position vector, and θ is the angle between $\hat{\mathbf{i}}$ and $\mathbf{r}$. Since the unit vectors $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\theta}}$ depend on θ which changes with time, the unit vectors also change with time. We will begin by looking at the derivative of $\hat{\mathbf{r}}$ with respect to time.

$$\frac{d\hat{\mathbf{r}}}{dt} = \frac{d\theta}{dt} \frac{d}{d\theta} \hat{\mathbf{r}} = \frac{d\theta}{dt} [-\hat{\mathbf{i}} \sin(\theta) + \hat{\mathbf{j}} \cos(\theta)] = \frac{d\theta}{dt} \hat{\boldsymbol{\theta}} \quad (1.3)$$

where we have used equation 1.2 to replace the typical Cartesian coordinate unit vectors.

$$\frac{d\hat{\boldsymbol{\theta}}}{dt} = \frac{d\theta}{dt} \frac{d}{dt} \hat{\boldsymbol{\theta}} = \frac{d\theta}{dt} [-\hat{\mathbf{i}} \cos(\theta) - \hat{\mathbf{j}} \sin(\theta)] = -\frac{d\theta}{dt} \hat{\mathbf{r}} \quad (1.4)$$

Since $\frac{d\hat{\mathbf{r}}}{dt}$ too depends on θ so is also time dependent.

$$\frac{d^2 \hat{\mathbf{r}}}{dt^2} = \frac{d}{dt} \left[\frac{d\theta}{dt} \hat{\boldsymbol{\theta}} \right] = \frac{d^2 \theta}{dt^2} \hat{\boldsymbol{\theta}} + \frac{d\theta}{dt} \frac{d\hat{\boldsymbol{\theta}}}{dt} = \frac{d^2 \theta}{dt^2} \hat{\boldsymbol{\theta}} - \left(\frac{d\theta}{dt} \right)^2 \hat{\mathbf{r}} \quad (1.5)$$

Polar unit vectors summary

The polar coordinate unit vectors are

$$\begin{aligned} \hat{\mathbf{r}} &= \hat{\mathbf{i}} \cos(\theta) + \hat{\mathbf{j}} \sin(\theta) \\ \hat{\boldsymbol{\theta}} &= -\hat{\mathbf{i}} \sin(\theta) + \hat{\mathbf{j}} \cos(\theta) \end{aligned}$$

With the first derivatives being

$$\begin{aligned} \frac{d\hat{\mathbf{r}}}{dt} &= \omega \hat{\boldsymbol{\theta}} \\ \frac{d\hat{\boldsymbol{\theta}}}{dt} &= -\omega \hat{\mathbf{r}} \end{aligned}$$

And the second derivative of $\hat{\mathbf{r}}$ being

$$\frac{d^2 \hat{\mathbf{r}}}{dt^2} = \frac{d\omega}{dt} \hat{\boldsymbol{\theta}} - \omega^2 \hat{\mathbf{r}}$$

with $\omega = \frac{d\theta}{dt}$

1.2 Velocity and acceleration in polar coordinates

Here we will calculate the acceleration in terms of the polar unit vectors.

1.2.1 Position

Starting with the position vector

$$\mathbf{r} = r\hat{\mathbf{r}} \quad (1.6)$$

This is the length of the position vector in the direction of $\hat{\mathbf{r}}$.

1.2.2 Velocity

The first derivative will be the velocity of the position vector in terms of the polar unit vectors

$$\frac{d\mathbf{r}}{dt} = \frac{dr}{dt}\hat{\mathbf{r}} + r\frac{d\hat{\mathbf{r}}}{dt} = \frac{dr}{dt}\hat{\mathbf{r}} + r\omega\hat{\boldsymbol{\theta}} \quad (1.7)$$

The velocity is the sum of the radial velocity in the radial direction $\dot{r}$ and the angular part of the velocity $r\omega$ in the angular direction.

1.2.3 Acceleration

$$\begin{aligned} \frac{d^2\mathbf{r}}{dt^2} &= \frac{d^2r}{dt^2}\hat{\mathbf{r}} + \frac{dr}{dt}\frac{d\hat{\mathbf{r}}}{dt} + \frac{dr}{dt}\omega\hat{\boldsymbol{\theta}} + r\frac{d\omega}{dt}\hat{\boldsymbol{\theta}} + r\omega\frac{d\hat{\boldsymbol{\theta}}}{dt} \\ &= \frac{d^2r}{dt^2}\hat{\mathbf{r}} + \frac{dr}{dt}\omega\hat{\boldsymbol{\theta}} + \frac{dr}{dt}\omega\hat{\boldsymbol{\theta}} + r\frac{d\omega}{dt}\hat{\boldsymbol{\theta}} - r\omega^2\hat{\mathbf{r}} \\ &= \left[\frac{d^2r}{dt^2} - r\omega^2 \right] \hat{\mathbf{r}} + \left[2\frac{dr}{dt}\omega + r\frac{d\omega}{dt} \right] \hat{\boldsymbol{\theta}} \end{aligned} \quad (1.8)$$

Acceleration in polar coordinates summary

The position vector

$$\mathbf{r} = r\hat{\mathbf{r}}$$

Velocity

$$\frac{d\mathbf{r}}{dt} = \frac{dr}{dt}\hat{\mathbf{r}} + r\omega\hat{\boldsymbol{\theta}}$$

Acceleration

$$\frac{d^2\mathbf{r}}{dt^2} = \left[\frac{d^2r}{dt^2} - r\omega^2 \right] \hat{\mathbf{r}} + \left[2\frac{dr}{dt}\omega + r\frac{d\omega}{dt} \right] \hat{\boldsymbol{\theta}}$$

2 Gravity

2.1 Gravity as a central force

The force between two particles a distance r apart with mass M and m respectively is given by

$$\mathbf{F} = -\frac{GMm}{r^2}\hat{\mathbf{r}} \quad (2.1)$$

The acceleration on the particle of mass m due to the force of gravity between the two particles is

$$\ddot{\mathbf{r}} = \frac{\mathbf{F}}{m} = -\frac{GM}{r^2}\hat{\mathbf{r}} \quad (2.2)$$

Using equation 1.8 we can split the acceleration into radial and angular parts

$$-\frac{GM}{r^2} = \ddot{r} - r\omega^2 \quad (2.3)$$

$$0 = r\ddot{\theta} + 2\dot{r}\dot{\theta} \quad (2.4)$$

3 Angular Momentum

The angular momentum of a particle is given by

$$\begin{aligned}\mathbf{L} &= \mathbf{r} \times \mathbf{p} \\ &= m\mathbf{r} \times \frac{d\mathbf{r}}{dt} \\ &= mr\hat{\mathbf{r}} \times \left[\dot{r}\hat{\mathbf{r}} + r\omega\hat{\boldsymbol{\theta}} \right] \\ &= m \left[r\dot{r}\hat{\mathbf{r}} \times \hat{\mathbf{r}} + r^2\omega\hat{\mathbf{r}} \times \hat{\boldsymbol{\theta}} \right] \\ &= mr^2\omega\hat{\mathbf{k}}\end{aligned}\tag{3.1}$$

We can also write this as $\mathbf{L} = l\hat{\mathbf{k}}$ where

$$l = mr^2\omega\tag{3.2}$$

3.1 Conservation Of Angular Momentum

We can show now that angular momentum is conserved under a central force by taking the derivative with respect to time.

$$\frac{d\mathbf{L}}{dt} = \frac{dl\hat{\mathbf{k}}}{dt} = \frac{dl}{dt}\hat{\mathbf{k}}\tag{3.3}$$

$$\begin{aligned}\frac{dl}{dt} &= m\frac{d}{dt} [r^2\omega] \\ &= m \left[\frac{dr^2}{dt}\omega + r^2\frac{d\omega}{dt} \right] \\ &= m \left[\frac{dr^2}{dr}\frac{dr}{dt}\omega + r^2\frac{d\omega}{dt} \right] \\ &= m \left[2r\frac{dr}{dt}\omega + r^2\frac{d\omega}{dt} \right] \\ &= mr \left[2\frac{dr}{dt}\omega + r\frac{d\omega}{dt} \right]\end{aligned}\tag{3.4}$$

The part inside the brackets is the angular acceleration which for a central force is zero¹ and so

$$\begin{aligned}\frac{dl}{dt} &= mr \times 0 \\ &= 0\end{aligned}\tag{3.5}$$

Which means that the angular momentum is constant.

¹see equation 2.4

4 Energy

4.1 Potential Energy

The potential energy is the work done in moving the particle from a radius of ∞ to r . So the work done is given by

$$\begin{aligned}U(r) &= \int_{\infty}^r \mathbf{F}(r') \cdot d\mathbf{r}' \\U(r) &= \int_{\infty}^r -\frac{GMm}{r'^2} \hat{\mathbf{r}} \cdot dr' \hat{\mathbf{r}} \\U(r) &= -GMm \int_{\infty}^r \frac{dr'}{r'^2} \\U(r) &= -GMm \left[\frac{-1}{r'} \right]_{\infty}^r \\U(r) &= GMm \left[\frac{1}{\infty} - \frac{1}{r} \right] \\U(r) &= -\frac{GMm}{r}\end{aligned}\tag{4.1}$$

4.2 Kinetic Energy

The kinetic energy of a particle is given by

$$T(\dot{\mathbf{r}}) = \frac{1}{2} m \dot{\mathbf{r}} \cdot \dot{\mathbf{r}}\tag{4.2}$$

using equation 1.2.3 gives

$$\begin{aligned}T(\dot{\mathbf{r}}) &= \frac{1}{2} m \left[\dot{r} \hat{\mathbf{r}} + r\omega \hat{\boldsymbol{\theta}} \right] \cdot \left[\dot{r} \hat{\mathbf{r}} + r\omega \hat{\boldsymbol{\theta}} \right] \\&= \frac{1}{2} m \dot{r}^2 + \frac{1}{2} m r^2 \omega^2\end{aligned}\tag{4.3}$$

we can also make the substitution using using the magnitude of the angular momentum² $l = mr^2\omega$

²equation 3.2

$$\begin{aligned}
l &= mr^2\omega \\
\omega &= \frac{l}{mr^2}
\end{aligned} \tag{4.4}$$

using equation 4.4 in equation 4.3 leads to

$$\begin{aligned}
T(\dot{r}) &= \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\omega^2 \\
&= \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2 \frac{l^2}{m^2r^4} \\
&= \frac{1}{2}m\dot{r}^2 + \frac{l^2}{2mr^2}
\end{aligned} \tag{4.5}$$

4.2.1 Total Energy

Finally the total energy can be found by putting together the equation for kinetic energy and potential energy

$$\begin{aligned}
E(r, \dot{r}) &= T(\dot{r}) + U(r) \\
E(r, \dot{r}) &= \frac{1}{2}m\dot{r}^2 + \frac{l^2}{2mr^2} - \frac{GMm}{r}
\end{aligned} \tag{4.6}$$

sometimes the last two terms of equation 4.6 are put together to form a new potential energy type term that is dependent only on the radius and the angular momentum.

4.3 Conservation of Total Energy

Just like with angular momentum we can show that total energy is conserved by taking the derivative with respect to time.

$$\begin{aligned}
\frac{dE}{dt} &= \frac{d}{dt} \left[\frac{1}{2} m \dot{r}^2 + \frac{l^2}{2mr^2} - \frac{GMm}{r} \right] \\
&= \frac{1}{2} m \frac{d\dot{r}^2}{dt} + \frac{l^2}{2m} \frac{dr^{-2}}{dt} - GMm \frac{dr^{-1}}{dt} \\
&= \frac{1}{2} m \frac{d\dot{r}^2}{d\dot{r}} \frac{d\dot{r}}{dt} + \frac{l^2}{2m} \frac{dr^{-2}}{dr} \frac{dr}{dt} - GMm \frac{dr^{-1}}{dr} \frac{dr}{dt} \\
&= \frac{1}{2} m (2\dot{r}) \ddot{r} + \frac{l^2}{2m} \left(\frac{-2}{r^3} \right) \dot{r} + \frac{GMm}{r^2} \dot{r} \\
&= m\dot{r}\ddot{r} - \frac{l^2}{mr^3} \dot{r} + \frac{GMm}{r^2} \dot{r}
\end{aligned} \tag{4.7}$$

using the magnitude of the angular momentum³ $l = mr^2\omega$

$$\begin{aligned}
\frac{dE}{dt} &= m\dot{r}\ddot{r} - \frac{l^2}{mr^3} \dot{r} + \frac{GMm}{r^2} \dot{r} \\
&= m\dot{r}\ddot{r} - \frac{m^2 r^4 \omega^2}{mr^3} \dot{r} + \frac{GMm}{r^2} \dot{r} \\
&= m\dot{r}\ddot{r} - mr\omega^2 \dot{r} + \frac{GMm}{r^2} \dot{r} \\
&= m\dot{r} \left[\ddot{r} - r\omega^2 + \frac{GM}{r^2} \right]
\end{aligned} \tag{4.8}$$

using the radial part of the acceleration⁴ $-\frac{GM}{r^2} = \ddot{r} - r\omega^2$ we get

$$\begin{aligned}
\frac{dE}{dt} &= m\dot{r} \left[-\frac{GM}{r^2} + \frac{GM}{r^2} \right] \\
&= m\dot{r} [0] \\
&= 0
\end{aligned} \tag{4.9}$$

This then shows that the total energy does not change with respect to time, so is a constant and also conserved.

³equation 3.2

⁴equation 2.3

5 Elliptical Orbits

5.1 Terminology

TODO a section about what periapsis and apoapsis, plus other terms. Maybe just a section deriving things about ellipses here?

5.2 Solving equations of motion

Firstly we let $r = \frac{1}{u}$, so that

$$\begin{aligned}\frac{dr}{dt} &= \frac{du^{-1}}{dt} \\ &= \frac{du^{-1}}{du} \frac{du}{dt} \\ &= -\frac{1}{u^2} \frac{du}{dt} \\ &= -r^2 \frac{du}{dt}\end{aligned}\tag{5.1}$$

next we introduce an ω by chain rule

$$\begin{aligned}\frac{dr}{dt} &= -r^2 \frac{du}{dt} \\ \frac{dr}{dt} &= -r^2 \frac{du}{d\theta} \frac{d\theta}{dt} \\ &= -r^2 \omega \frac{du}{d\theta}\end{aligned}$$

And using the magnitude of angular momentum⁵ $l = mr^2\omega$ or rearranging $\omega = \frac{l}{mr^2}$

⁵equation 3.2

$$\begin{aligned}
\frac{dr}{dt} &= -r^2\omega \frac{du}{d\theta} \\
\frac{dr}{dt} &= -r^2 \frac{l}{mr^2} \frac{du}{d\theta} \\
\frac{dr}{dt} &= -\frac{l}{m} \frac{du}{d\theta} \\
\frac{dr}{dt} &= -\frac{l}{m} \frac{du}{d\theta}
\end{aligned} \tag{5.2}$$

Differentiating again with respect to time gives

$$\begin{aligned}
\frac{dr^2}{dt^2} &= -\frac{l}{m} \frac{d}{dt} \frac{du}{d\theta} \\
&= -\frac{l}{m} \frac{d\theta}{dt} \frac{d}{d\theta} \frac{du}{d\theta} \\
&= -\frac{l}{m} \omega \frac{d^2u}{d\theta^2}
\end{aligned} \tag{5.3}$$

We can substitute the magnitude of angular momentum⁶ $l = mr^2\omega$ again to get

$$\begin{aligned}
\frac{dr^2}{dt^2} &= -\frac{l}{m} \omega \frac{d^2u}{d\theta^2} \\
&= -\frac{mr^2\omega}{m} \omega \frac{d^2u}{d\theta^2} \\
&= -r^2\omega^2 \frac{d^2u}{d\theta^2}
\end{aligned} \tag{5.4}$$

We can now substitute this into the radial part of the acceleration⁷

⁶equation 3.2

⁷equation 2.3

$$\begin{aligned}
-\frac{GM}{r^2} &= \frac{d^2 r}{dt^2} - r\omega^2 \\
&= -r^2\omega^2 \frac{d^2 u}{d\theta^2} - r\omega^2 \\
&= -r^2\omega^2 \left[\frac{d^2 u}{d\theta^2} + \frac{1}{r} \right] \\
GM &= r^4\omega^2 \left[\frac{d^2 u}{d\theta^2} + \frac{1}{r} \right]
\end{aligned} \tag{5.5}$$

Next we recall that $u = \frac{l}{r}$ and that $\frac{l}{m} = r^2\omega$ and substitute these in

$$\begin{aligned}
GM &= \left(\frac{l}{m} \right)^2 \left[\frac{d^2 u}{d\theta^2} + u \right] \\
\frac{d^2 u}{d\theta^2} + u &= \frac{GMm^2}{l^2}
\end{aligned} \tag{5.6}$$

To simplify this equation we will use $\alpha = \frac{GMm^2}{l^2}$

5.2.1 Solution

The direct solution to

$$\frac{d^2 u}{d\theta^2} + u = \alpha \tag{5.7}$$

is $u = \alpha$

However because there is also a function that results in zero, we must include that

$$\begin{aligned}
\frac{d^2 u}{d\theta^2} + u &= 0 \\
\frac{d^2 u}{d\theta^2} &= -u
\end{aligned} \tag{5.8}$$

we can take the solution that $u = c \cos \theta$ which gives the general overall solution of

$$u(\theta) = \alpha + c \cos \theta \quad (5.9)$$

We can pick an initial condition when $\theta = 0$ the body is at periapsis or $r = p = A(1 - e)$ which gives $u = \frac{1}{A(1-e)}$ (see the ellipse notes for $p = A(1 - e)$)

$$\begin{aligned} \frac{1}{A(1 - e)} &= \alpha + c \\ \alpha &= \frac{1}{A(1 - e)} - c \end{aligned} \quad (5.10)$$

we could similarly have picked the condition that when $\theta = \pi$ the body is at apoapsis or $r = a = A(1 + e)$ which gives $u = \frac{1}{A(1+e)}$ (see ellipse notes for $a = A(1 + e)$)

$$\begin{aligned} \frac{1}{A(1 + e)} &= \alpha - c \\ \alpha &= \frac{1}{A(1 + e)} + c \end{aligned} \quad (5.11)$$

we can equate our two initial conditions to get c in terms of just a and p

$$\begin{aligned}
\alpha &= \frac{1}{A(1-e)} - c = \frac{1}{A(1+e)} + c \\
\frac{1}{A(1-e)} - \frac{1}{A(1+e)} &= 2c \\
\frac{A(1+e) - A(1-e)}{A^2(1+e)(1-e)} &= 2c \\
\frac{A + eA - A + eA}{A^2(1-e^2)} &= 2c \\
\frac{2eA}{A^2(1-e^2)} &= 2c \\
\frac{e}{A(1-e^2)} &= c
\end{aligned} \tag{5.12}$$

Now that we have c we can use this to determine α in terms of A and e

$$\begin{aligned}
\alpha &= \frac{1}{A(1+e)} + c \\
&= \frac{1}{A(1+e)} + \frac{e}{A(1-e)^2} \\
&= \frac{(1-e)}{A(1+e)(1-e)} + \frac{e}{A(1-e)^2} \\
&= \frac{1-e+e}{A(1-e^2)} \\
&= \frac{1}{A(1-e^2)}
\end{aligned}$$

So feeding all this back in we have

$$\begin{aligned}
u(\theta) &= \alpha + c \cos \theta \\
&= \frac{1}{A(1 - e^2)} + \frac{e}{A(1 - e^2)} \cos \theta \\
&= \frac{1 + e \cos \theta}{A(1 - e^2)}
\end{aligned} \tag{5.13}$$

Now we have a simple form for u we can recall that we chose $u = \frac{1}{r}$, so

$$r = \frac{1 + e \cos \theta}{A(1 - e^2)} \tag{5.14}$$

which is the equation for a ellipse we saw from the ellipse notes. We also get the interesting result here that

$$\begin{aligned}
\alpha &= \frac{GMm^2}{l^2} = \frac{1}{A(1 - e^2)} \\
A(1 - e^2) &= \frac{l^2}{GMm^2}
\end{aligned} \tag{5.15}$$